

# Power Transmission Component Market Top Trends & Analysis 2035

The power transmission component market forms the backbone of modern electricity networks, enabling efficient transfer of power from generation sources to end users. In 2025, the global market size was valued at over USD 81.6 billion and is projected to reach USD 155.1 billion by the end of 2035. The market is expected to expand at a compound annual growth rate (CAGR) of 7.4% during the forecast period from 2026 to 2035. This robust expansion is driven by rising electricity demand, grid modernization initiatives, renewable energy integration, and cross-border transmission projects.

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## Power Transmission Component Industry Demand

Power transmission components include critical equipment such as transformers, transmission lines, conductors, circuit breakers, switchgear, insulators, surge arrestors, and transmission towers. These components collectively ensure the safe, reliable, and efficient transmission of electricity across long distances and varied terrains.

Industry demand for power transmission components is increasing due to rapid urbanization, industrial growth, electrification of transportation, and the expansion of renewable power capacity. Utilities and grid operators favor these components for their long operational life, high reliability, and cost-effective performance over extended periods. Advances in material science and manufacturing have improved durability and reduced maintenance requirements, while standardized designs and modular systems simplify installation and system administration. The ability of modern components to withstand high voltages, extreme weather, and fluctuating loads further strengthens market demand.

## Power Transmission Component Market: Growth Drivers & Key Restraint

### Growth Drivers

- **Expansion and Modernization of Power Grids**  
Aging transmission infrastructure in many regions is being replaced or upgraded to improve grid stability, reduce transmission losses, and support smart grid deployment. This has significantly increased demand for advanced transmission components.
- **Rising Integration of Renewable Energy Sources**  
Wind, solar, and hydropower projects require long-distance, high-capacity transmission networks. This has accelerated the adoption of high-voltage transmission lines, efficient transformers, and advanced switchgear solutions.

- **Technological Advancements and Cost Optimization**

Innovations such as high-voltage direct current systems, advanced insulating materials, compact switchgear, and high-temperature conductors have enhanced efficiency while lowering lifecycle costs, making large-scale deployment more economically viable.

## **Restraint**

- **High Capital Investment and Regulatory Complexity**

Large-scale transmission projects involve significant upfront investment, long approval timelines, and strict regulatory requirements, which can delay project execution and limit short-term market growth.

## **Power Transmission Component Market: Segment Analysis**

### **Segment Analysis by Transformers**

Transformers across voltage ranges play a vital role in voltage regulation and power flow management. Lower-voltage transformers are widely used in regional distribution networks, while higher-voltage transformers support long-distance bulk power transmission. Demand is strongest for transformers that offer higher efficiency, compact design, and improved thermal performance.

### **Segment Analysis by Transmission Lines and Conductors**

High-voltage AC transmission remains widely deployed for regional networks, while high-voltage DC systems are increasingly preferred for long-distance and cross-border transmission due to lower losses. High-temperature low-sag conductors are gaining traction as utilities seek to increase line capacity without constructing new corridors. Traditional aluminum conductor steel-reinforced solutions continue to be used for their mechanical strength and proven performance.

### **Segment Analysis by Circuit Breakers and Switchgear**

Gas-insulated switchgear is widely adopted in space-constrained urban and industrial environments, while air-insulated switchgear remains common in open substations. Vacuum circuit breakers are increasingly used due to their safety and low maintenance. Growing environmental concerns are driving the shift toward SF6 alternatives such as G<sup>3</sup> and dry air technologies.

### **Segment Analysis by Insulators**

Composite and polymer insulators are gaining preference due to their lightweight design, pollution resistance, and reduced maintenance needs. Porcelain and glass insulators continue to be used in conventional transmission networks due to their mechanical strength and long service history.

## **Segment Analysis by Surge Arrestors**

Metal-oxide varistor arrestors dominate due to their superior protection against transient overvoltages. Gapped silicon carbide arrestors and line surge arresters are used in specific grid configurations to enhance system reliability and protect critical infrastructure.

## **Segment Analysis by Transmission Towers and Structures**

Lattice steel towers remain the most widely deployed due to structural strength and adaptability. Tubular steel poles are increasingly used in urban and coastal areas for aesthetic and corrosion-resistance benefits. Hybrid and composite structures are emerging as utilities seek lightweight, durable, and environmentally resilient solutions. Guyed towers continue to be used where cost efficiency and terrain conditions permit.

## **Power Transmission Component Market: Regional Insights**

### **North America**

The North American market is driven by grid refurbishment programs, renewable energy integration, and rising electricity consumption. Investments in smart grids, high-voltage transmission corridors, and resilience against extreme weather events are key demand drivers.

### **Europe**

Europe's market growth is supported by cross-border power interconnections, offshore wind expansion, and strict efficiency and sustainability regulations. The region places strong emphasis on environmentally friendly switchgear and advanced transmission technologies.

### **Asia-Pacific (APAC)**

APAC represents the fastest-growing regional market due to rapid industrialization, urban expansion, and large-scale electrification projects. Government-led investments in power infrastructure and renewable energy capacity are driving strong demand for transmission components.

## **Top Players in the Power Transmission Component Market**

The power transmission component market features a diverse mix of global and regional players, including General Electric Company (U.S.), Eaton Corporation (U.S.), 3M Company (U.S.), Dow Inc. (U.S.), Siemens AG (Germany), BASF SE (Germany), Schneider Electric SE (France), ABB Ltd. (Switzerland), Arkema S.A. (France), Hitachi Chemical Co., Ltd. (Japan), Mitsubishi Electric Corporation (Japan), Sumitomo Electric Industries, Ltd. (Japan), LG Chem Ltd. (South Korea), Hyundai Electric & Energy Systems Co., Ltd. (South Korea), Tata Chemicals Limited (India), Reliance Industries Limited (India), Petronas Chemicals Group Berhad (Malaysia), Brambles Limited (Australia), Johnson Matthey Plc (UK), and Solvay S.A. (Belgium), all actively focusing on innovation, grid modernization, and global expansion strategies.

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Contact for more Info:

AJ Daniel

Email: [info@researchnester.com](mailto:info@researchnester.com)

U.S. Phone: +1 646 586 9123

U.K. Phone: +44 203 608 5919